

01

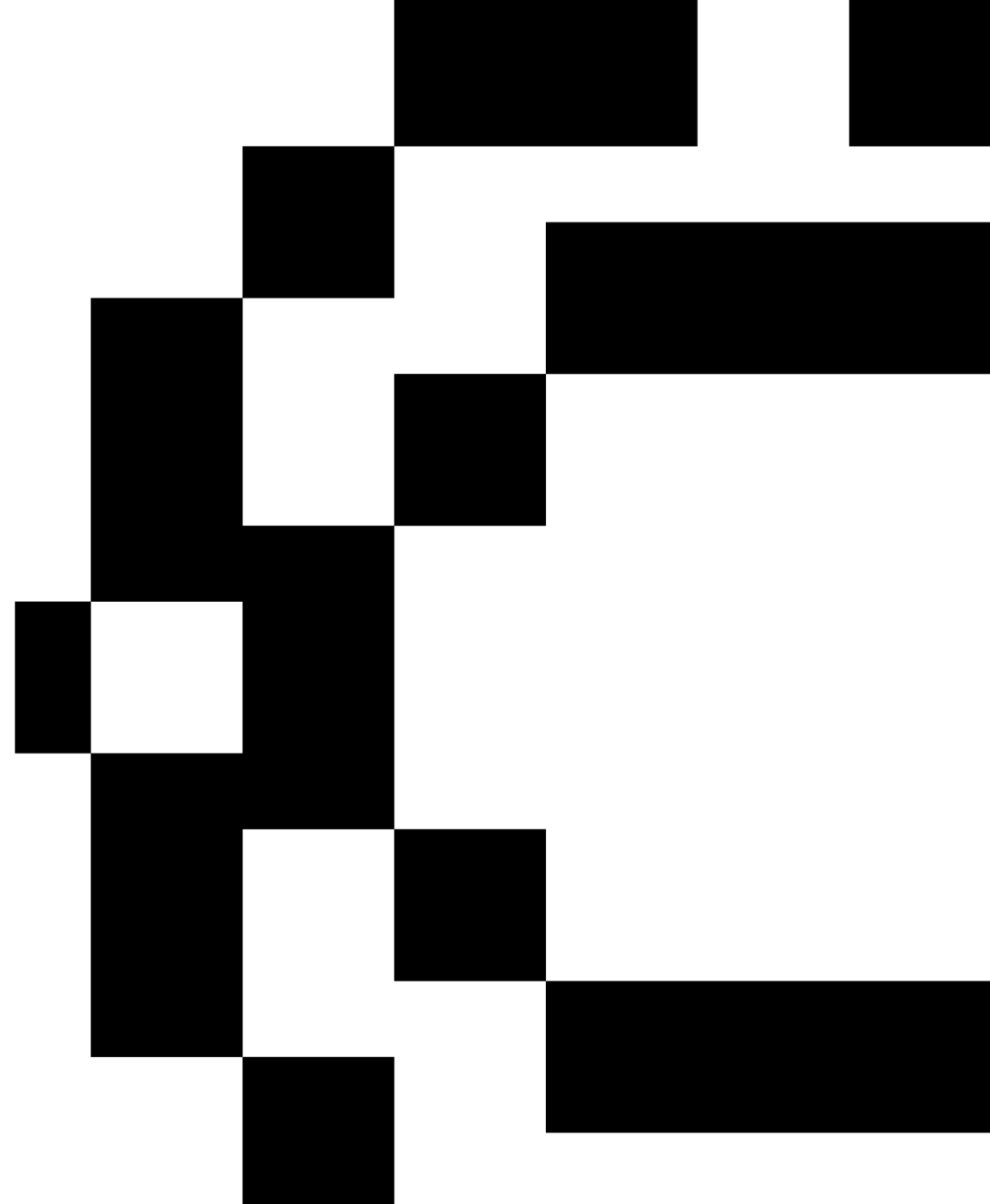
Introduction to Data Mining II

Data Mining II| MGI 2526

February - Week 1

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Agenda

01



02



03



04



05

**Introduction
to Supervised
Learning**

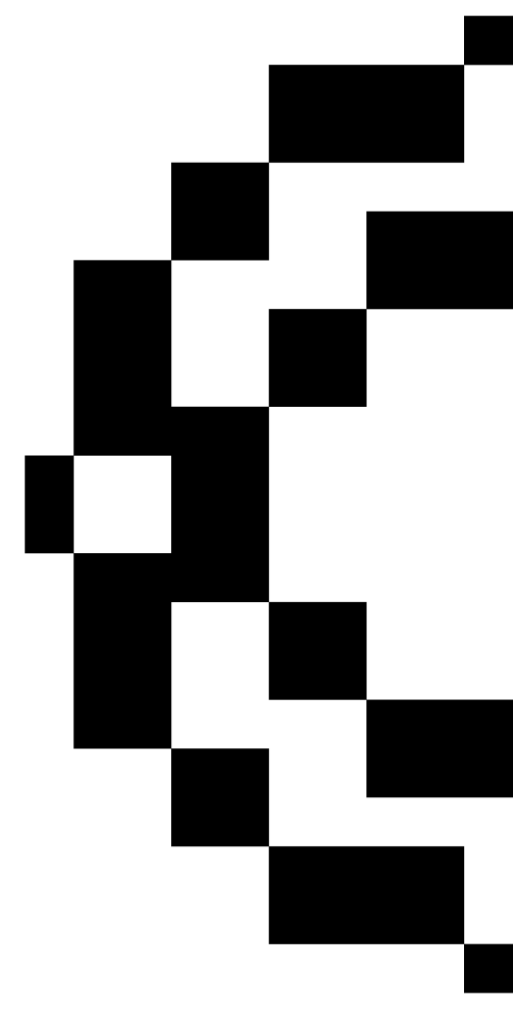
**Common
Misunderstandings**

**Methodologies
for Supervised
Learning**

**Ethical
Considerations**

**Challenges
and Future
Trends**

00. Quick Recap



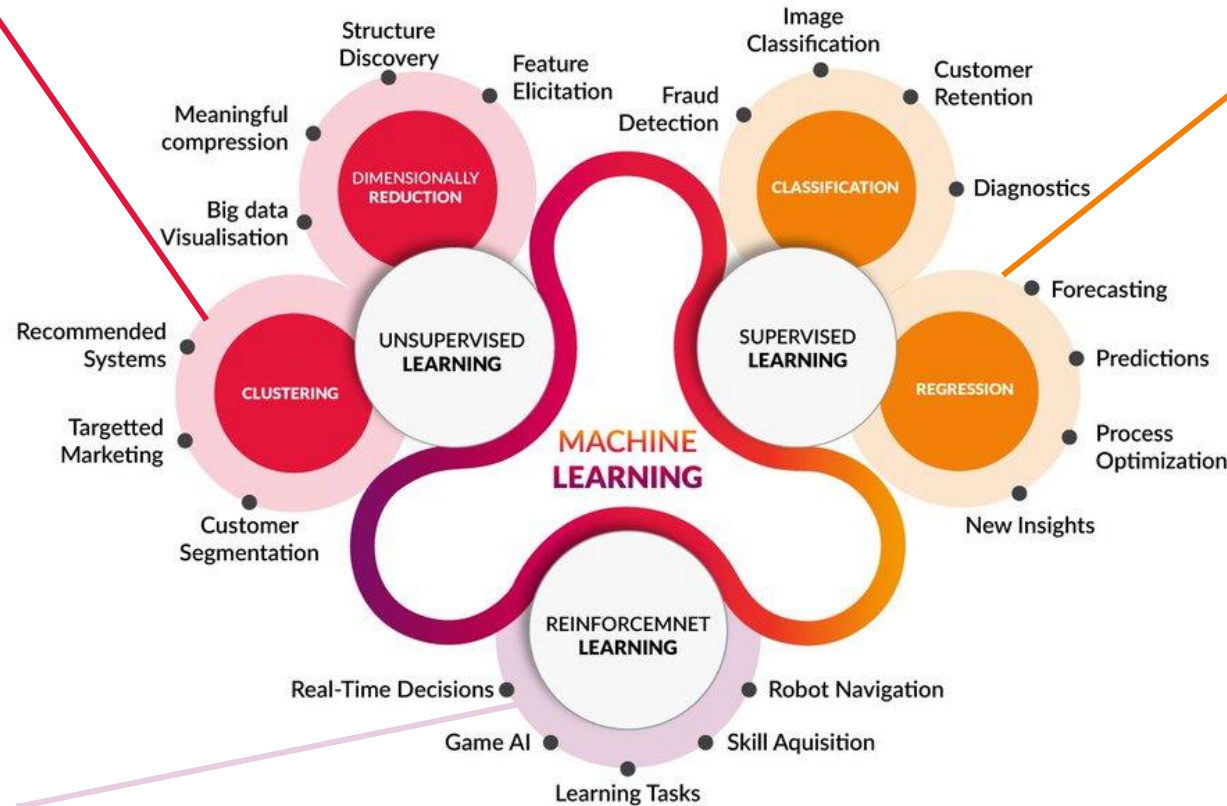
0.1. Supervised vs Unsupervised Learning

Introduction

The algorithm is trained on an **unlabeled dataset**.

Goal: identify patterns, relationships, and structures in the data, without being told what the target or output should be.

Data Mining I



The algorithm is trained on a **labeled dataset**, meaning that the desired output or target is provided for each example in the training data.

Goal: learn how to map inputs to outputs and to make predictions on new, unseen data.

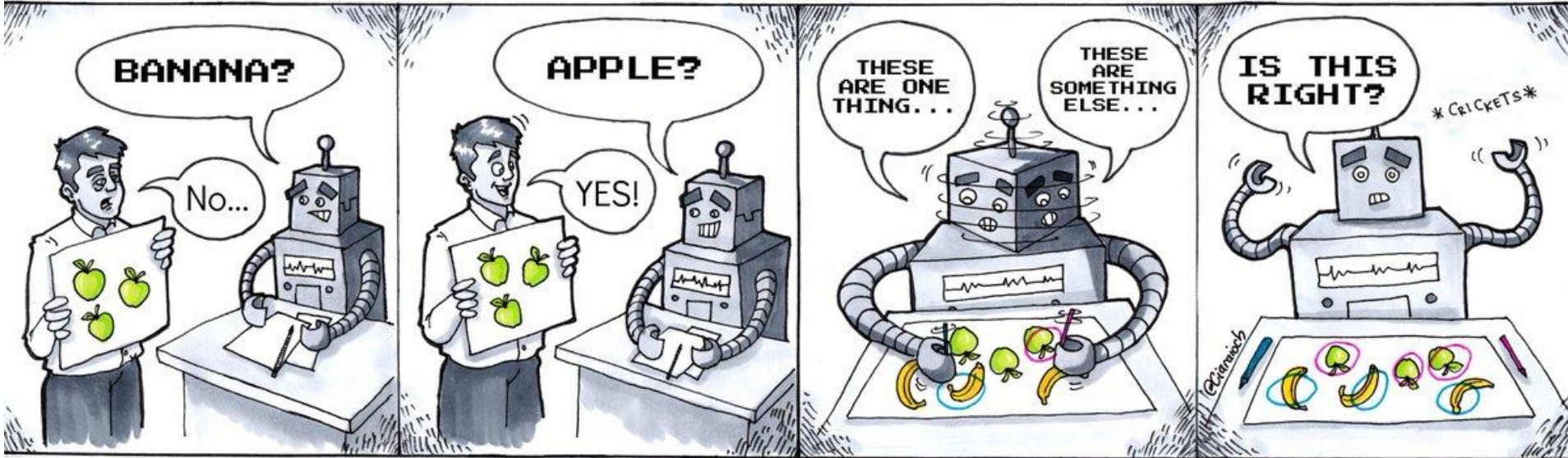
Data Mining II

The algorithm learns by taking actions in an environment and receiving rewards or penalties for those actions.

Goal: learn a policy, which is a mapping from states to actions, that maximizes the expected cumulative reward.

0.1. Supervised vs Unsupervised Learning

Introduction



Supervised Learning

- Training dataset with labels
- Learns relationship between features and target
- Used to classify unseen data points

Unsupervised Learning

- Data does not have labels
- Finds optimal clusters (groups)
- Not used with unseen data points

0.2. Supervised vs Unsupervised Learning

Similarities and Differences

What are the
similarities?

- Both are **types of machine learning**.
- Both use **algorithms to learn from data**.
- Both aim to **extract meaningful information** from datasets.
- Both can be used for **pattern recognition and decision-making**.
- Both may **require preprocessing** (cleaning, normalization, feature engineering).

0.2. Supervised vs Unsupervised Learning

Similarities and Differences

SO...

What are the
differences?

0.2. Supervised vs Unsupervised Learning

Similarities and Differences

	Unsupervised Learning	Supervised Learning
Definition	Learns from unlabeled data to discover hidden patterns	Learns from labeled data to predict outcomes
Type of Data	Unlabeled (input only)	Labeled (input + correct output / ground truth)
Main Goal	Identify structure or relationships in data	Predict a target value
Common tasks	Clustering, Dimensionality Reduction, Association	Classification, Regression

0.2. Supervised vs Unsupervised Learning

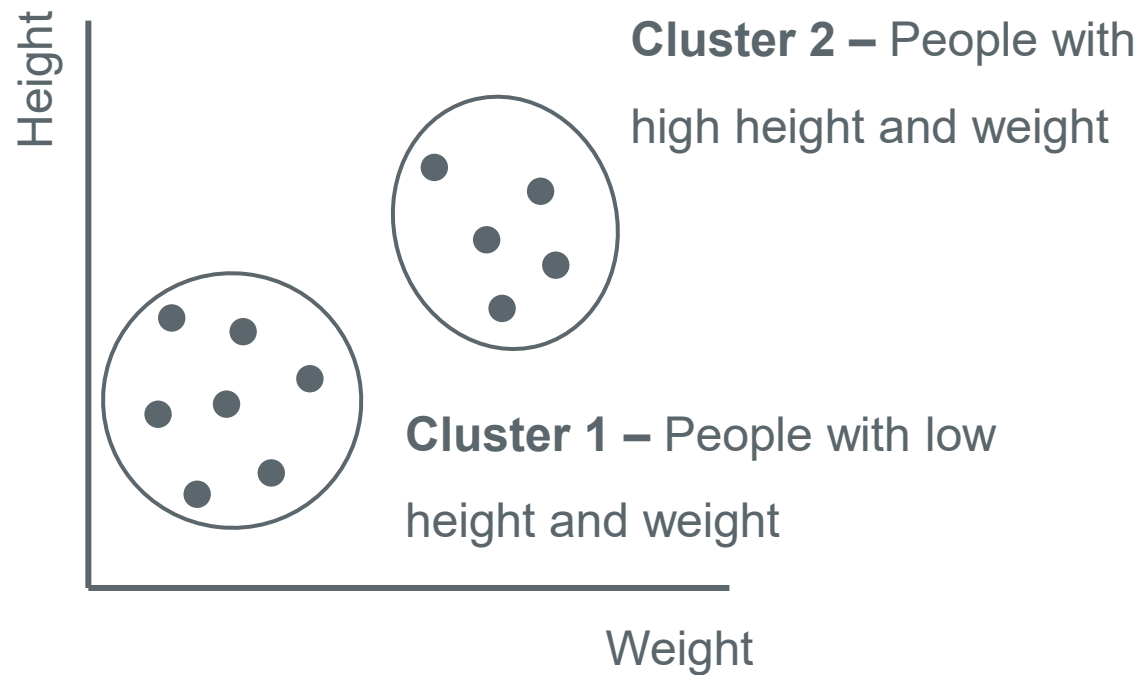
Similarities and Differences

	Unsupervised Learning	Supervised Learning
Algorithms (Example)	K-Means, Hierarchical Clustering, DBSCAN, Apriori, ...	Linear regression, Decision Tree, Neural Networks, ...
Output	Groups, clusters, patterns, or reduced feature space	Predicted labels or continuous values
Evaluation Methods	Silhouette score, inertia, cluster validity metrics, etc.	Accuracy, Precision, Recall, F1-score, RMSE, etc.
Complexity of Interpretation	Often more exploratory and harder to interpret	Usually easier to interpret (clear objective)

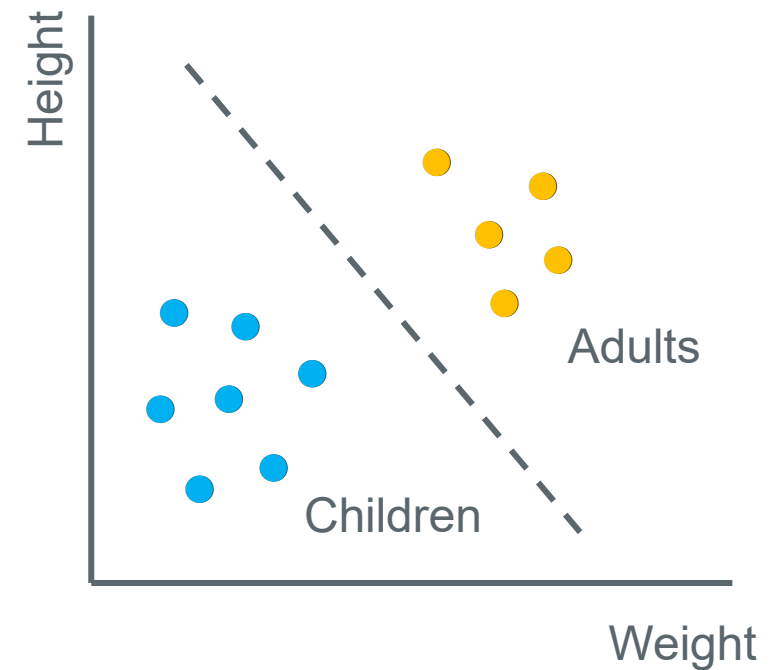
0.3. What is Unsupervised Learning?

Clustering vs Classification

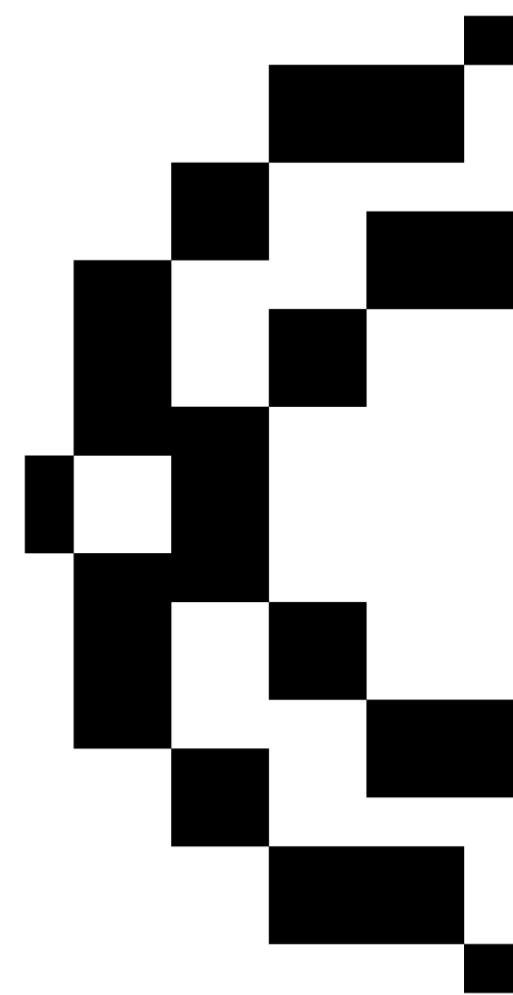
Clustering



Classification



01. Introduction to Supervised Learning



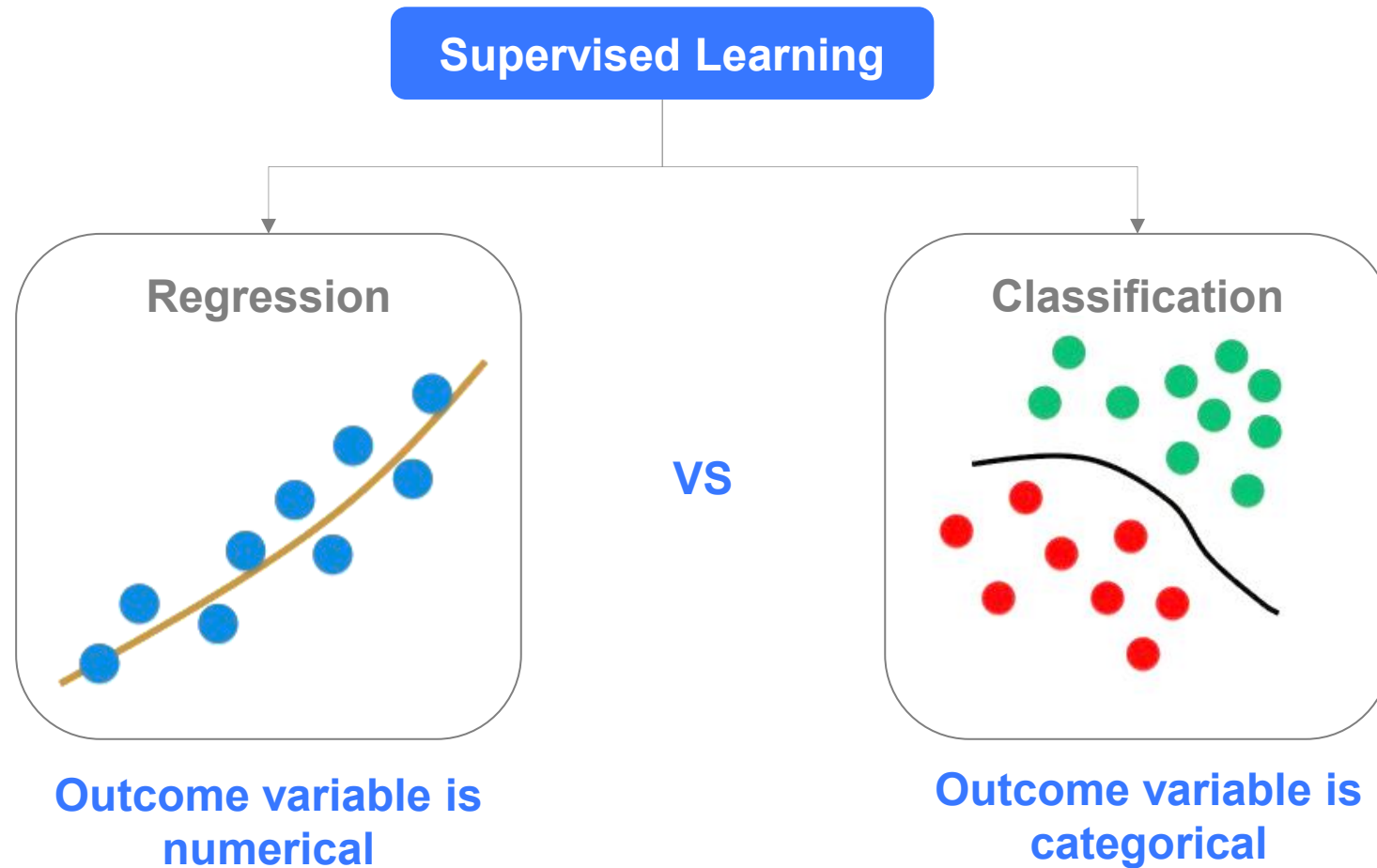
1. What is Supervised Learning?

Introduction

Machine learning techniques that use **labeled** datasets
to train algorithms with the purpose to **classify** data or **predict outcomes**

1. What is Supervised Learning?

Introduction



1.1. What is Supervised Learning?

Regression



Predicting the price of a house

Area	Typology	Localization	Garage	Pollution Index	Price
75	T2	Alfama	No	1	234 000
86	T1	Campolide	Yes	2	210 000
93	T2	Santos	No	2	254 000
134	T4	Arroios	No	3	457 000
96	T3	Alfama	No	1	367 000
54	T2	Alcântara	No	2	219 000
167	T5	Restelo	Yes	2	675 000
95	T2	Arroios	Yes	3	?

1.1. What is Supervised Learning?

Regression



Predicting stock market values

Day	Company	GDP acceleration	Inflation	...	Value
01	VLU	0.358	-1.059	...	20.17
01	RIT	0.121	0.205	...	125.23
01	XEZ	-0.121	-0.075	...	98.76
02	VLU	-0.256	0.172	...	20.19
02	RIT	0.232	1.212	...	145.87
02	XEZ	0.867	-0.754	...	95.23
03	VLU	...	...	...	?

Which variables proxy for those changes remains unknown ...

1.2. What is Supervised Learning?

Classification

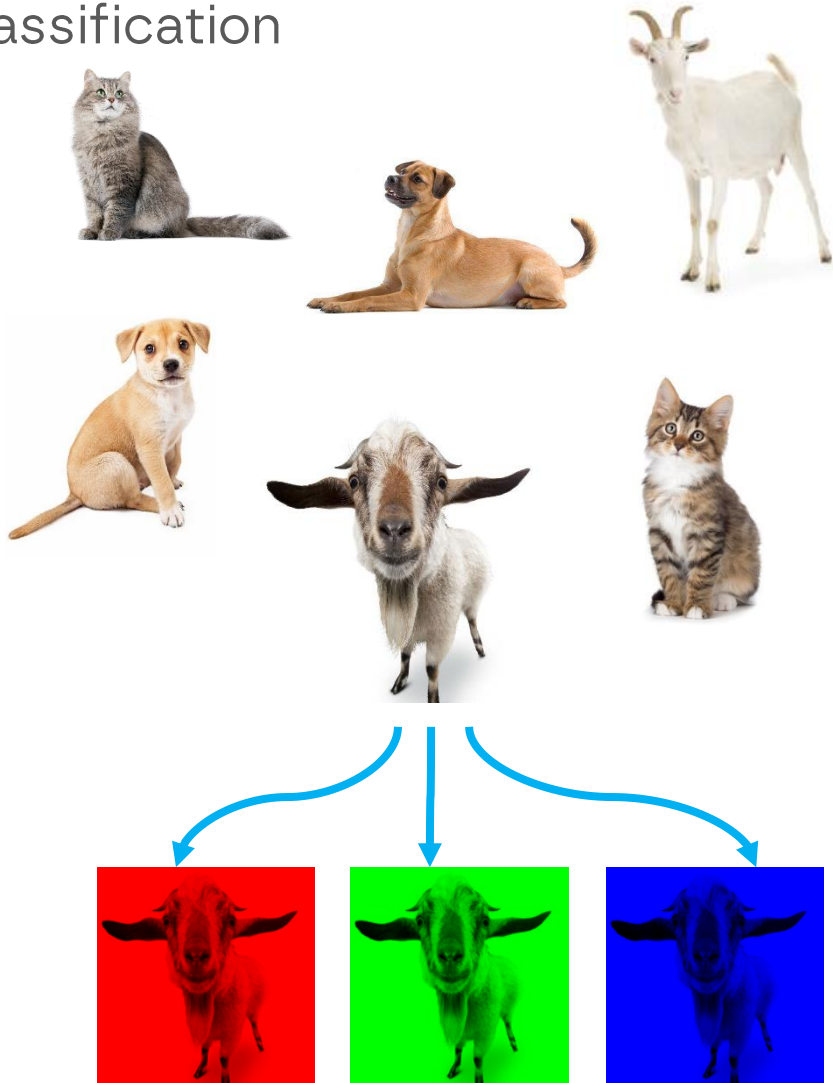


Predicting the probability of someone having diabetes

ID	Gender	Weight	Height	Pregnancies	Diabetes
12562	M	78	175	0	0
62735	F	66	155	3	1
84638	F	91	165	1	1
1232	M	89	187	0	0
13738	M	101	172	0	1
3274	M	81	179	0	0
3283	F	72	169	0	0
3627	F	93	185	0	?

1.2. What is Supervised Learning?

Classification



Predicting which animal is in a photo

Figure	Channel	Pixel_1	Pixel_2	Pixel_3	...	Animal
Fig_1	Red	126	134	136		Cat
Fig_1	Green	222	211	207		
Fig_1	Blue	198	198	196		
Fig_2	Red	35	31	31		Goat
Fig_2	Green	89	88	87		
Fig_2	Blue	145	147	149		
...	...	...	...	...	...	...

Figure	Channel	Pixel_1	Pixel_2	Pixel_3	...	Animal
Fig_156	Red	234	236	237		?
Fig_156	Green	176	176	175		
Fig_156	Blue	98	97	98		

1.3. What is Supervised Learning?

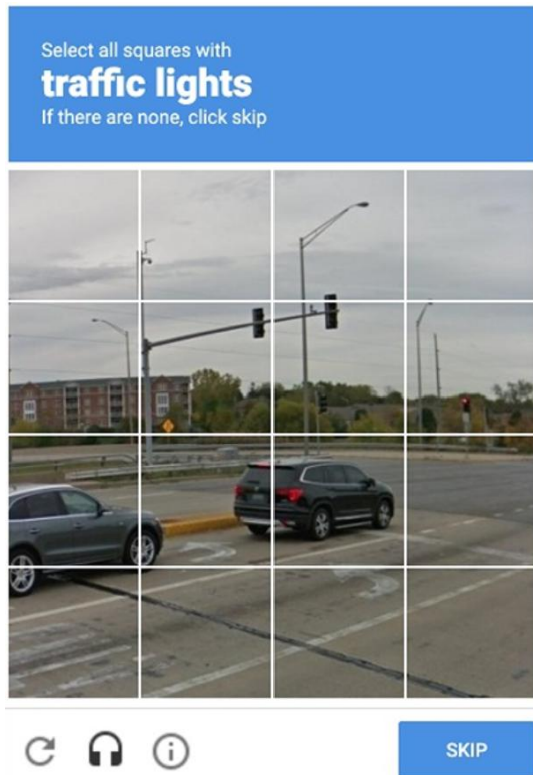
Regression + Classification

Can we have both **regression** and **classification** in a same model?

1.3. What is Supervised Learning?

Regression + Classification

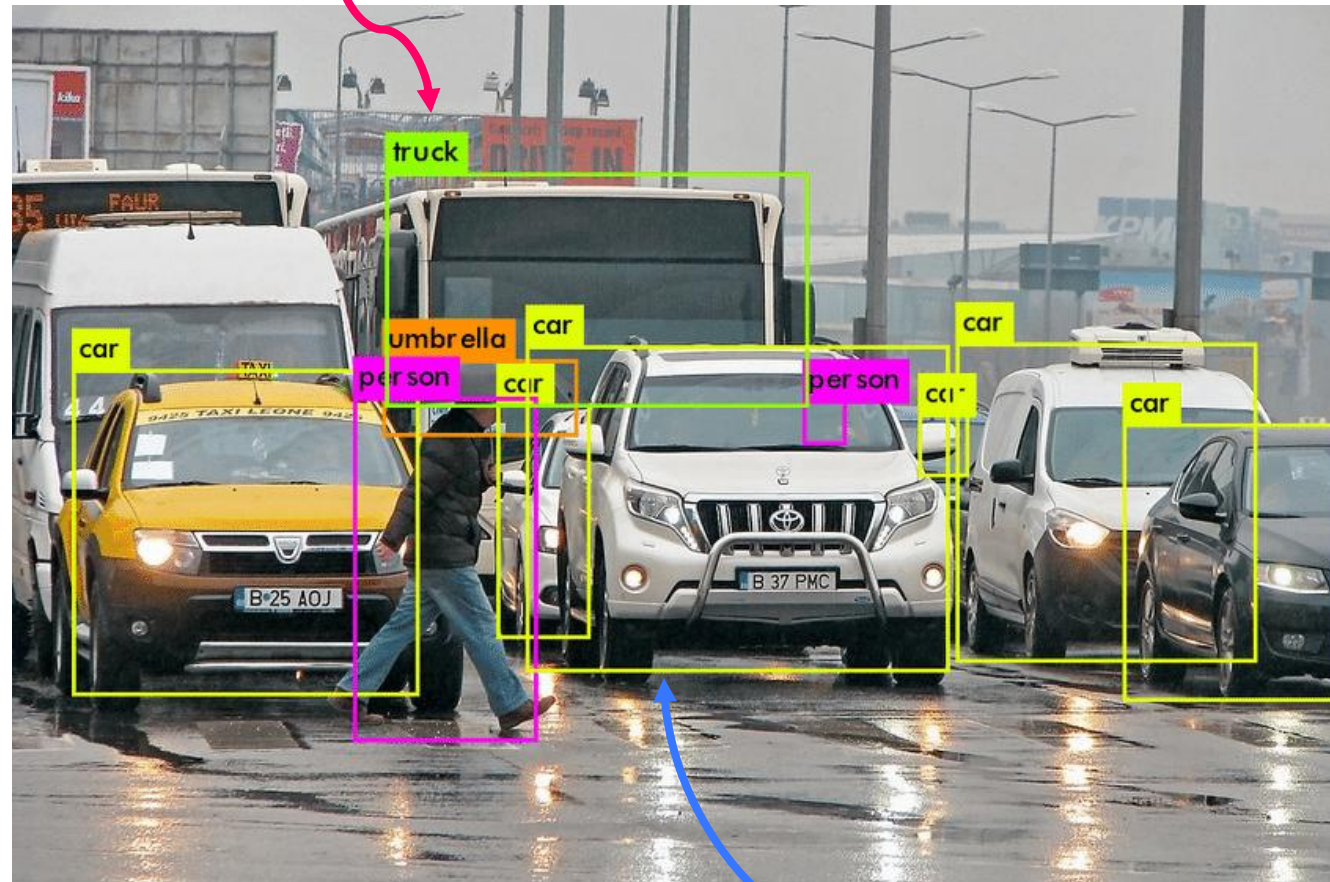
What does CAPTCHA tests have to do with automatic driving cars?



1.3. What is Supervised Learning?

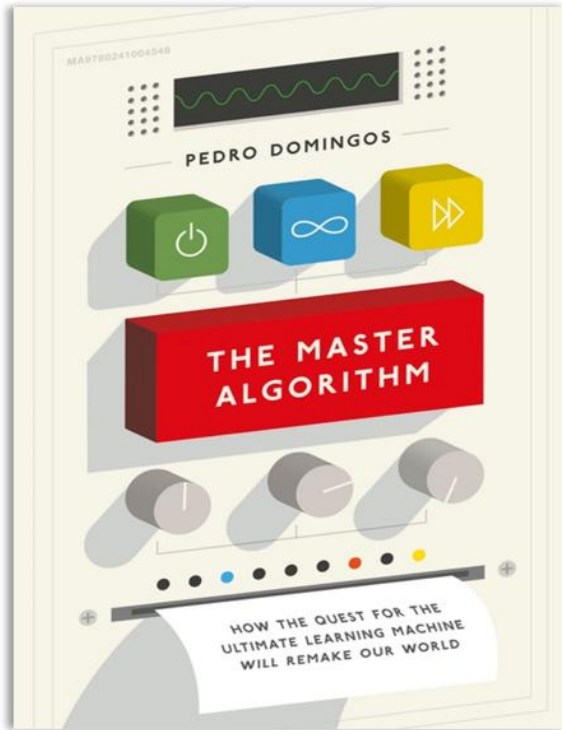
Regression + Classification

Classification (Class: Truck)

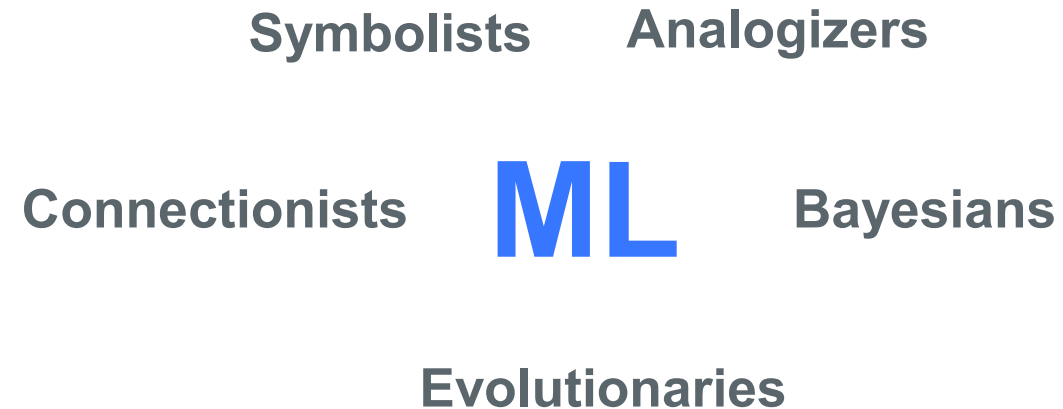


1.4. What is Supervised Learning?

The 5 tribes of Machine Learning



Domingos, P. (2015). *The master algorithm: How the quest for the ultimate learning machine will remake our world*. Basic Books.



1.4. What is Supervised Learning?

The 5 tribes of Machine Learning

Tribe	Origins	Master Algorithm
Symbolists	Logic, Philosophy	Inverse Deduction
Connectionists	Neuroscience	Backpropagation
Evolutionaries	Evolutionary Biology	Genetic Programming
Bayesians	Statistics	Probabilistic Inference
Analogizers	Psychology	Kernel Machines (SVM)

1.4.1. What is Supervised Learning?

The 5 tribes of Machine Learning

Symbolists

Origins: Logic, Philosophy

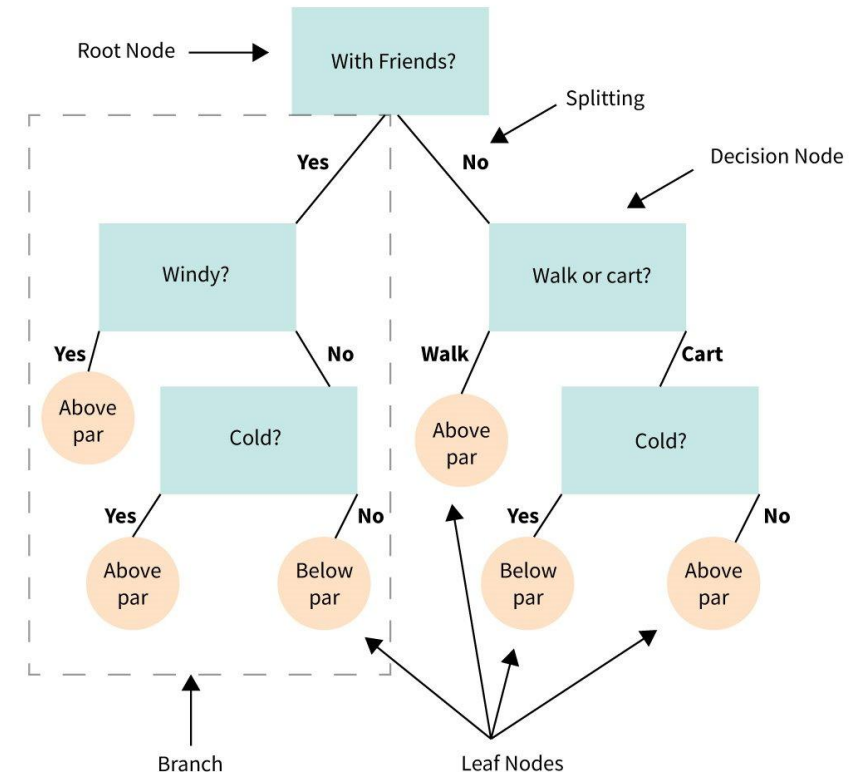
All intelligence can be reduced to manipulating symbols, in the same way that a mathematician solves equations by replacing expressions by other expressions.

Symbolists understand that you can't learn from scratch: you need some initial knowledge to go with the data.

They've figured out how to incorporate preexisting knowledge into learning, and how to combine different pieces of knowledge on the fly in order to solve new problems.

Master Algorithm: Inverse deduction

Figure out what knowledge is missing in order to make a deduction go through, and then makes it as general as possible.



In Data Mining II: Decision Trees

1.4.2. What is Supervised Learning?

The 5 tribes of Machine Learning

Connectionists

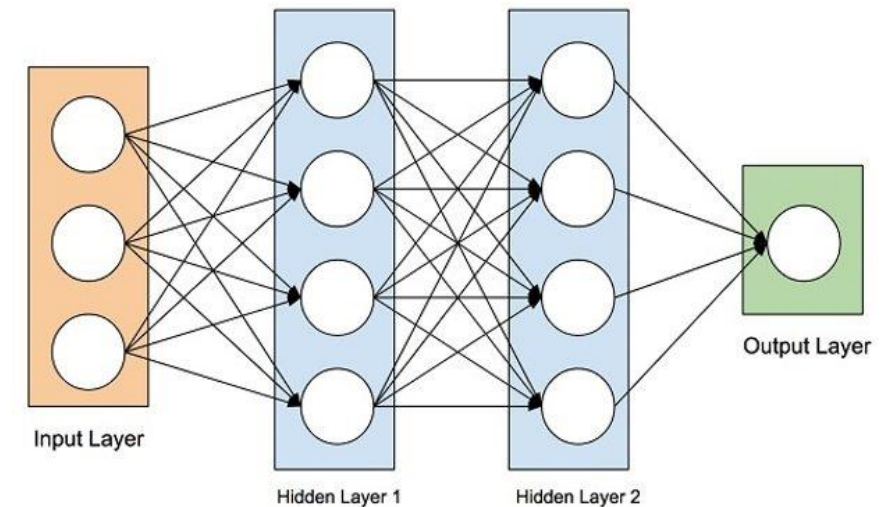
Origins: Neuroscience

Learning is what the brain does, and so what we need to do is reverse engineer it.

The brain learns by adjusting the strengths of connections between neurons, and the crucial problem is figuring out which connections are to blame for which errors and changing them accordingly.

Master Algorithm: Backpropagation

Compares a system's output with the desired one and then successively changes the connections in layer after layer of neurons so as to bring the output closer to what it should be.



In Data Mining II: Neural Networks

1.4.3. What is Supervised Learning?

The 5 tribes of Machine Learning

Evolutionaries

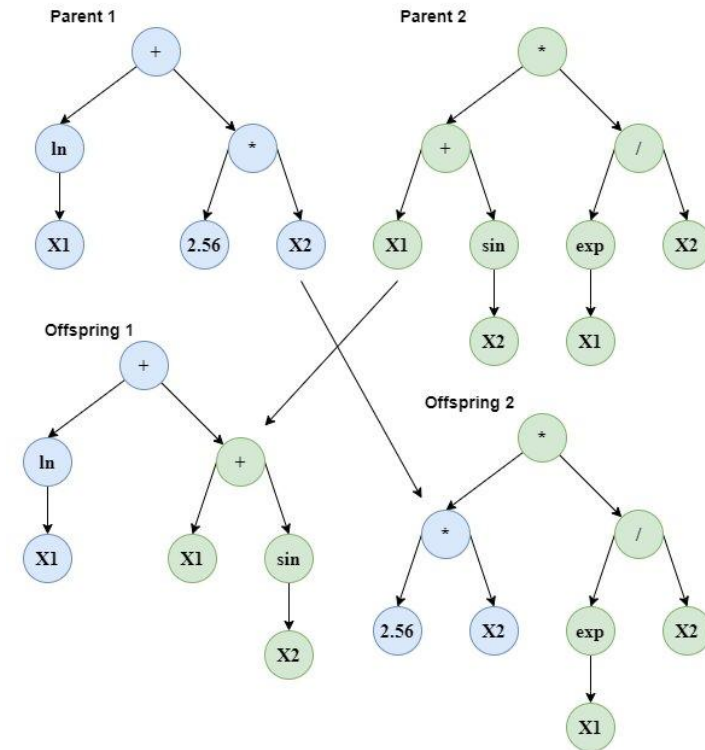
Origins: Evolutionary biology

Evolutionaries believe that the mother of all learning is natural selection. If it made us, it can make anything, and all we need to do is simulate it on the computer.

The key problem that evolutionaries solve is learning structure: not just adjusting parameters, like backpropagation does, but creating the brain that those adjustments can then fine-tune.

Master Algorithm: Genetic Programming

Mates and evolves computer programs in the same way that nature mates and evolves organisms.



In Data Mining II: Ensemble techniques

1.4.4. What is Supervised Learning?

The 5 tribes of Machine Learning

Bayesians

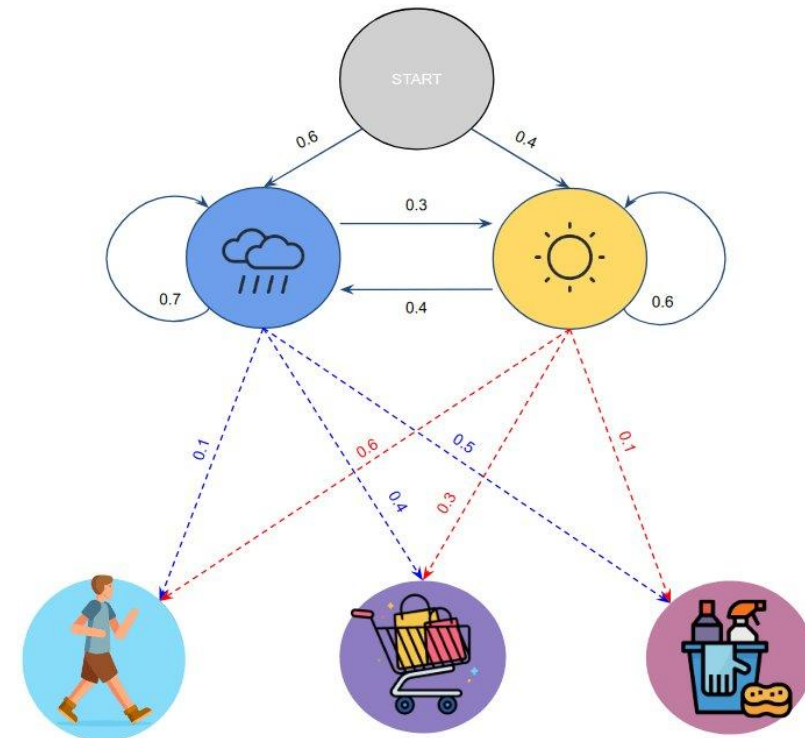
Origins: Statistics

Bayesians are concerned above all with uncertainty. All learned knowledge is uncertain, and learning itself is a form of uncertain inference.

The problem then becomes how to deal with noisy, incomplete, and even contradictory information without falling apart.

Master Algorithm: Bayes' Theorem

Tells us how to incorporate new evidence into our beliefs, and probabilistic inference algorithms do that as efficiently as possible.



In Data Mining II: Naïve Bayes

1.4.5. What is Supervised Learning?

The 5 tribes of Machine Learning

Analogizers

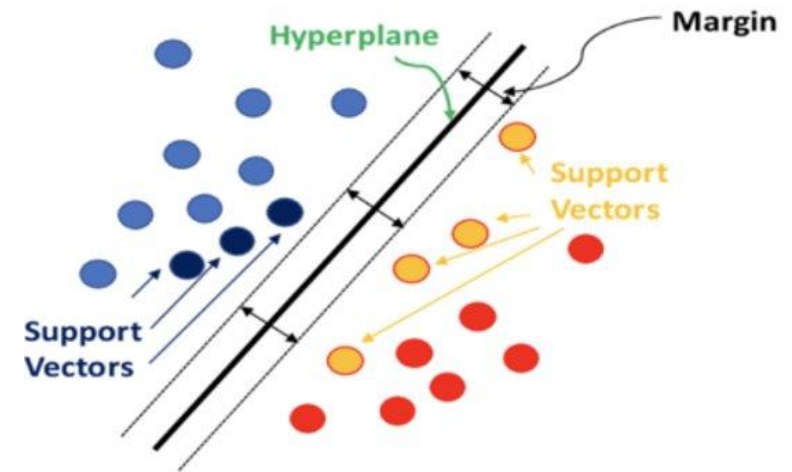
Origins: Psychology

The key to learning is recognizing similarities between situations and thereby inferring other similarities. If two patients have similar symptoms, perhaps they have the same disease.

The key problem is judging how similar two things are.

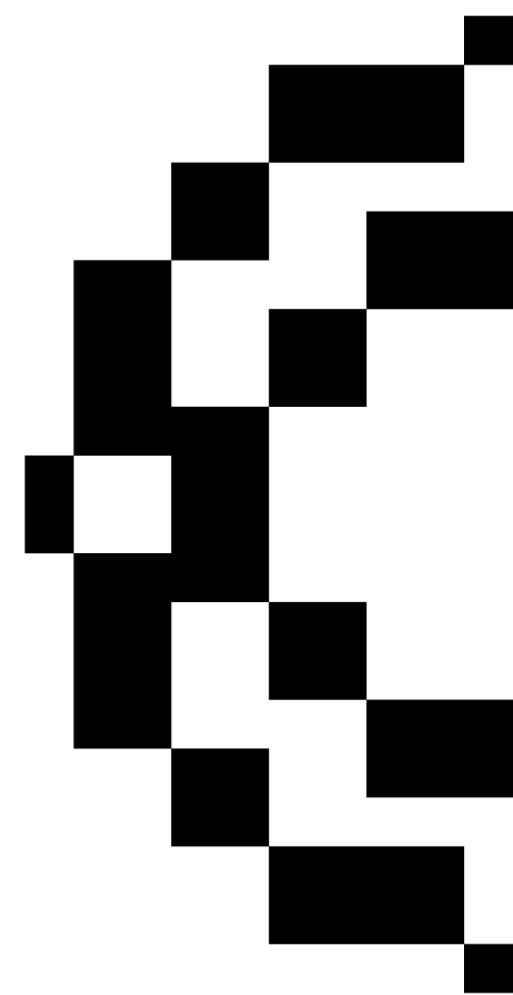
Master Algorithm: Kernel Machines (SVM)

Figures out which experiences to remember and how to combine them to make new predictions.



In Data Mining II: K-Nearest Neighbors

02. Common Misunderstandings



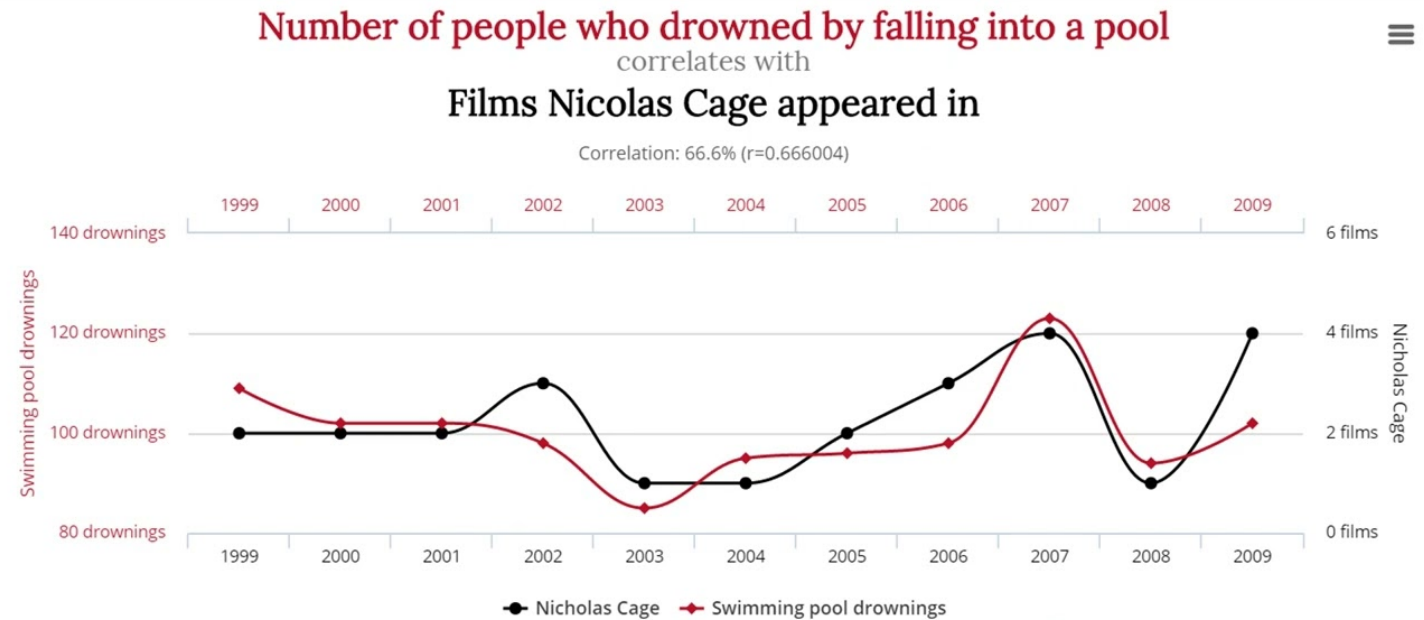
2.1. Common misunderstandings

Prediction vs Understanding

If the model is accurate,
it understands the
problem!

Reality: High accuracy means **good prediction**, not understanding.

A model can predict well using spurious correlations.



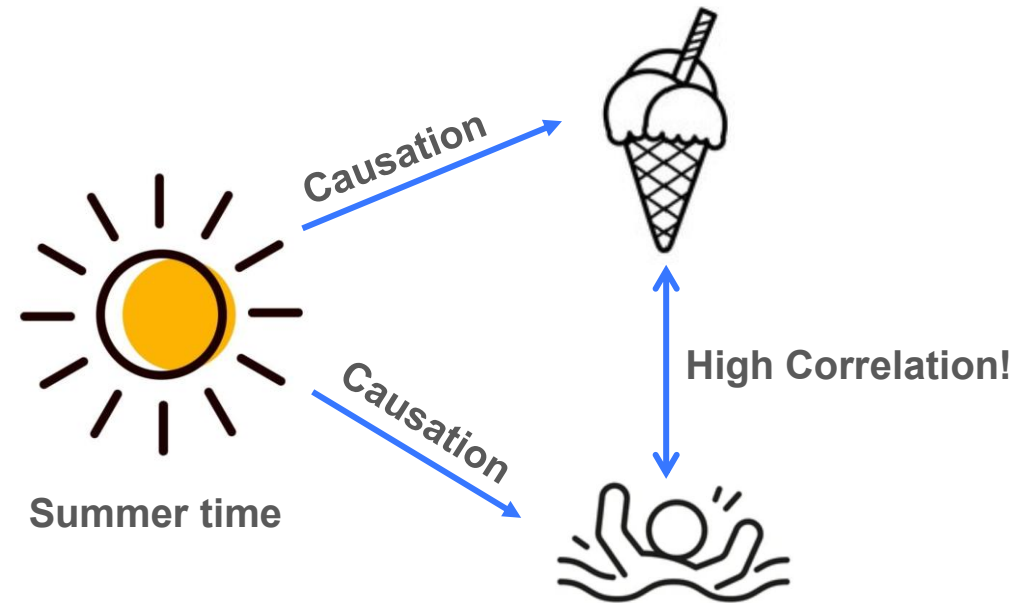
Data sources: Centers for Disease Control & Prevention and Internet Movie Database

2.2. Common misunderstandings

Causality vs Correlation

Reality: Supervised learning is usually **correlational**, not causal.
It answers *what will happen*, not *why it happens*.

Supervised Learning
finds the cause!

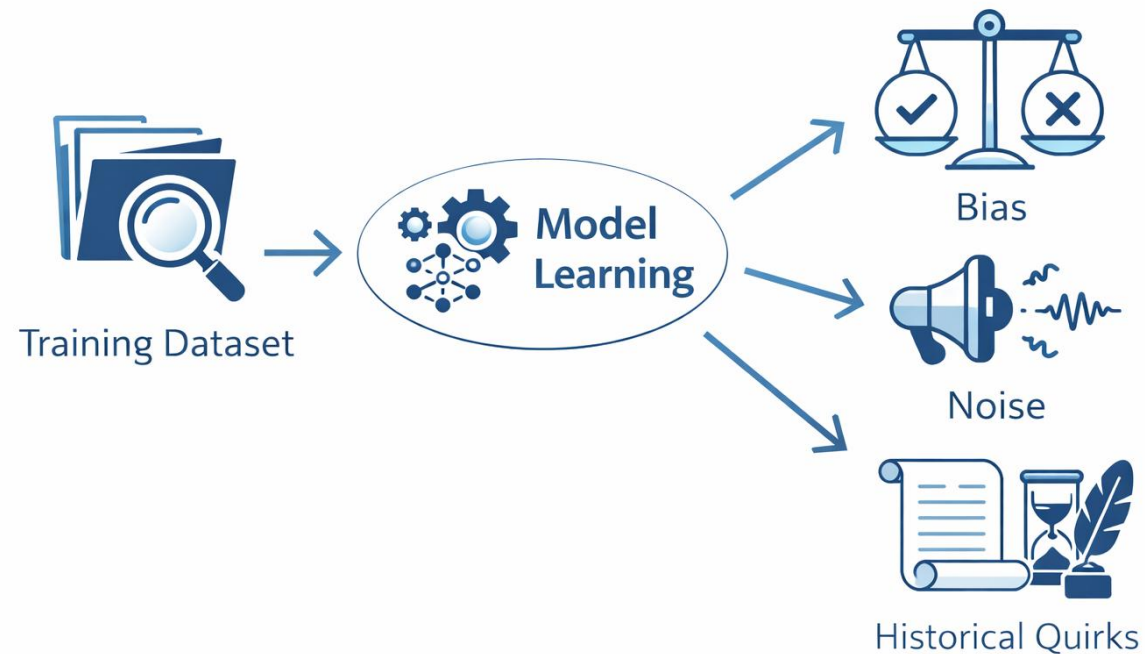


2.3. Common misunderstandings

Overall learning

The model learns the
rules of the world!

Reality: The model learns patterns in the **training dataset**, which may include **bias**, **noise**, and **historical quirks**.

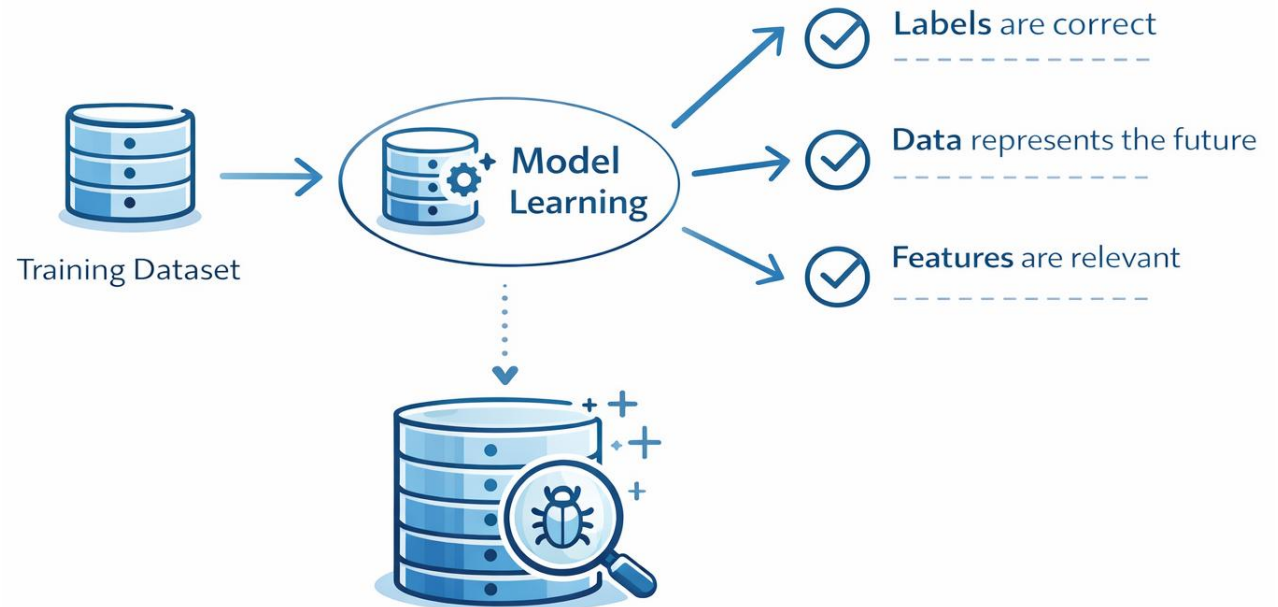


2.4. Common misunderstandings

More data, better performance

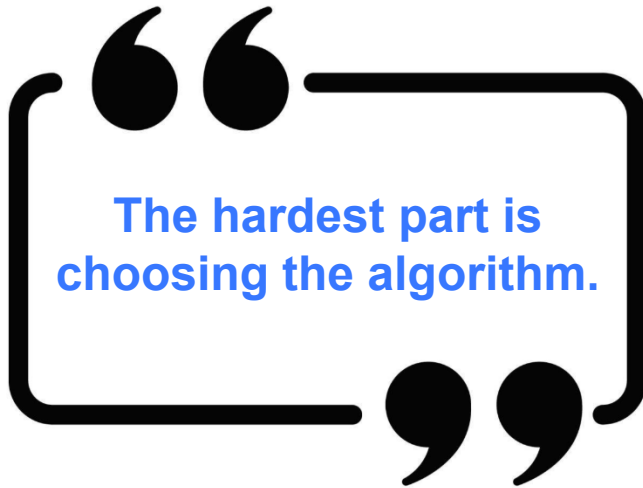
Reality: More data helps only if:

“
More data always means
better performance
”



2.5. Common misunderstandings

The hard is the algorithm



Reality: Often the hardest part is:

- defining the target properly
- collecting labels
- avoiding **leakage**
- evaluating correctly

The algorithm is rarely the main difficulty.

What is leakage? (more about it in future classes)

When your model gets access to information during training that it wouldn't have in real life when making predictions.

2.6. Common misunderstandings

Great accuracy, great model

Reality: It depends on:

- **class imbalance**
- error costs
- baseline performance



A model with 95% of accuracy is excellent!

Example: if 95% of customers don't churn, predicting "no churn" always gives 95% accuracy and is useless.

What is class imbalance? (more about it in future classes)

When the categories (classes) you're trying to predict are **not evenly represented** in your data.

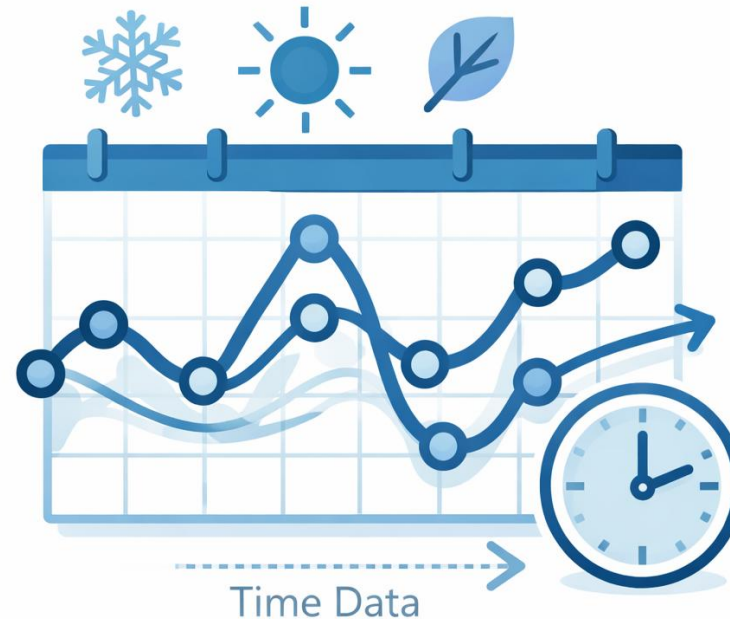
2.7. Common misunderstandings

Time series

Supervised Learning
works the same for
time-series data.

Reality: Time breaks key assumptions.

You can't randomly shuffle time data — and evaluation must respect chronological order.



2.8. Common misunderstandings

Ground truth



Reality: Labels are often:

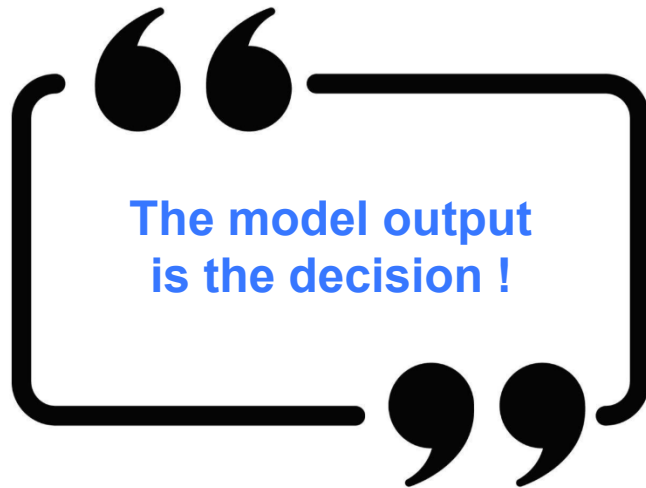
- noisy
- delayed
- subjective
- proxies for what we actually care about

The model can only learn what the labels represent.

2.9. Common misunderstandings

The output is the final decision

Reality: The model outputs a prediction. A decision requires:



- thresholds
- policies
- human judgment
- Ethics

Supervised learning **supports decision-making**,
it does not replace it!

2.10. Common Mistakes (Even in Thesis and Articles)

Prediction as a proof of causality

Most supervised ML models learn **associations**, not **causal influence**.



“Zip code influences loan approval.”

In reality, zip code may be a proxy for socioeconomic status, race, or location-based inequality.

→ For causality, you may need causal inference methods.

A better way to write it:

“Feature X was consistently among the most predictive variables for Y.”

2.11. Common Mistakes (Even in Thesis and Articles)

Accuracy without a baseline

Accuracy alone can be meaningless!



Problems:

- If the dataset is imbalanced, accuracy can be misleading.
- Without context, 92% might be terrible or great.
- No mention of baseline comparison.
- No confidence interval or variability.

A better way to write it:

“Our model achieved 92% accuracy on the test set; however, recall for the minority class was 61%.”

2.12. Common Mistakes (Even in Thesis and Articles)

Model training and evaluation

This sentence often hides a **serious methodological flaw**:



- training and testing on the *same data* → data leakage → inflated results
- unclear split method
- unclear whether preprocessing was done correctly

Also: “the dataset” is vague — what dataset, how many samples, what time period?

A better way to write it:

- “We split the dataset into training (70%), validation (15%), and test (15%) sets.”
- “Preprocessing steps were fitted on the training set only to avoid leakage.”
- “We used stratified 5-fold cross-validation.”

2.13. Common Mistakes (Even in Thesis and Articles)

Overgeneralization

This is an **overgeneralization**. It may work well:



- on your dataset
- for your population
- for your time period
- for your company/product type
- under your feature availability

But churn changes over time and across industries!

A better way to write it:

- “This approach performed well on our churn dataset compared to baselines.”
- “Further validation is required to assess generalization to other customer segments.”
- ...

2.14. Common Mistakes (Even in Thesis and Articles)

Deep learning is always better

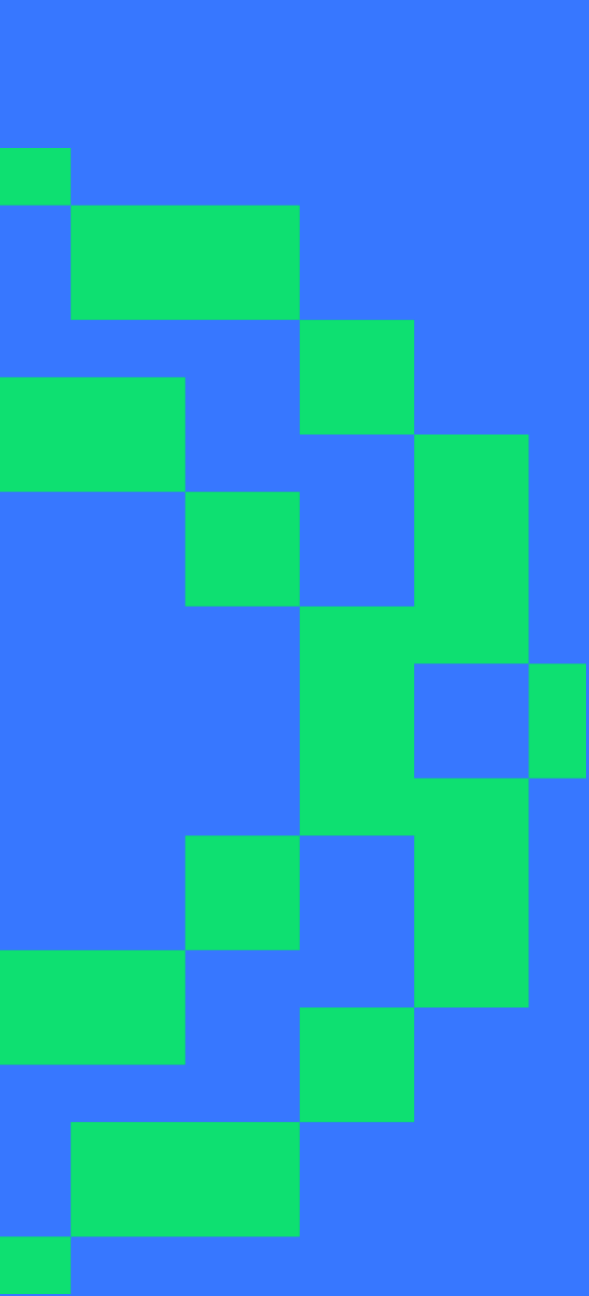


Deep learning is not always the best choice!

- Small datasets → classical models often perform better
- Tabular data → tree-based models (e.g., XGBoost) are often competitive
- High interpretability required → simpler models preferred
- Limited compute → deep learning may be impractical
- Marginal performance gains may not justify complexity

A better way to write it:

- “Deep learning achieved the highest F1-score, but required significantly more training time.”
- “XGBoost performed comparably to deep learning while being easier to interpret.”
- ...



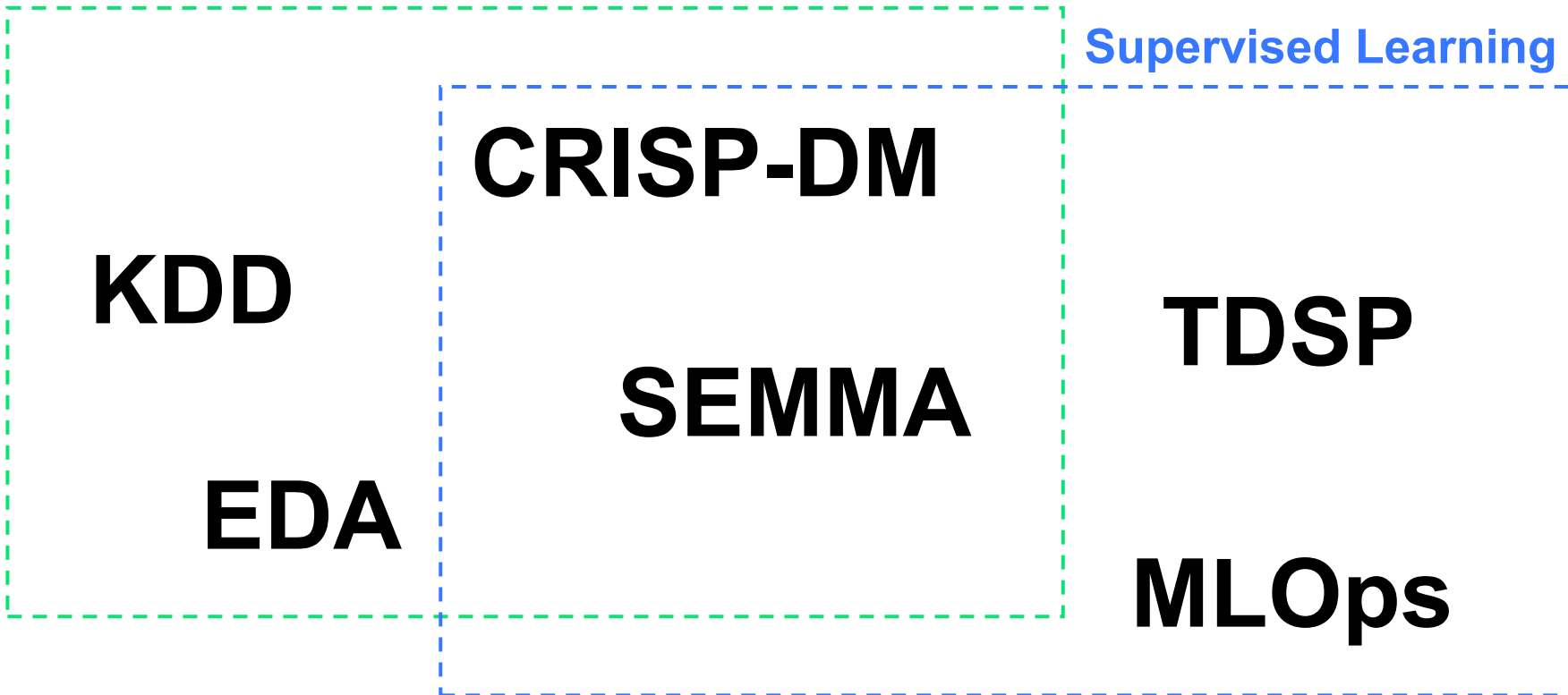
03.

Methodologies for Supervised Learning

3. Methodologies for Supervised Learning

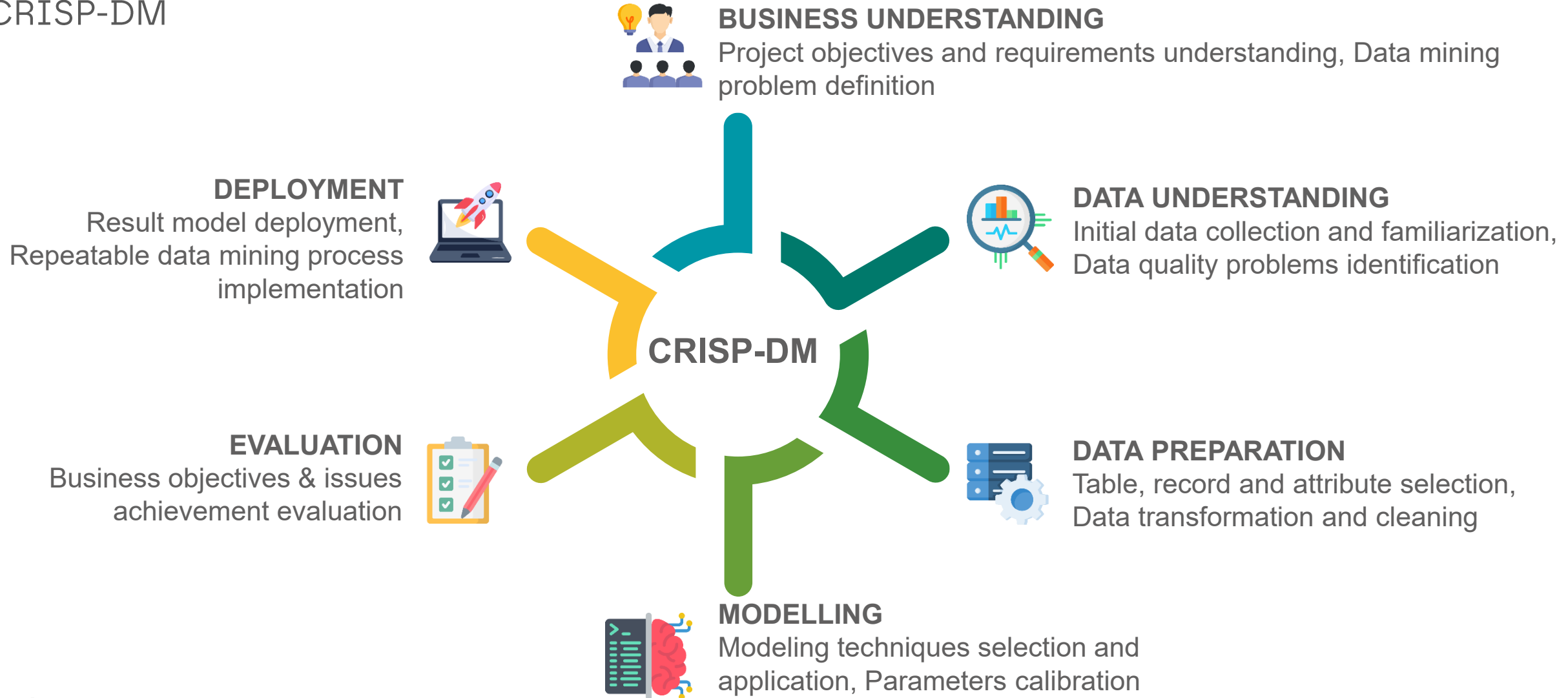
Introduction

Unsupervised Learning



3.1. Methodologies for Supervised Learning

CRISP-DM



3.2. Methodologies for Supervised Learning

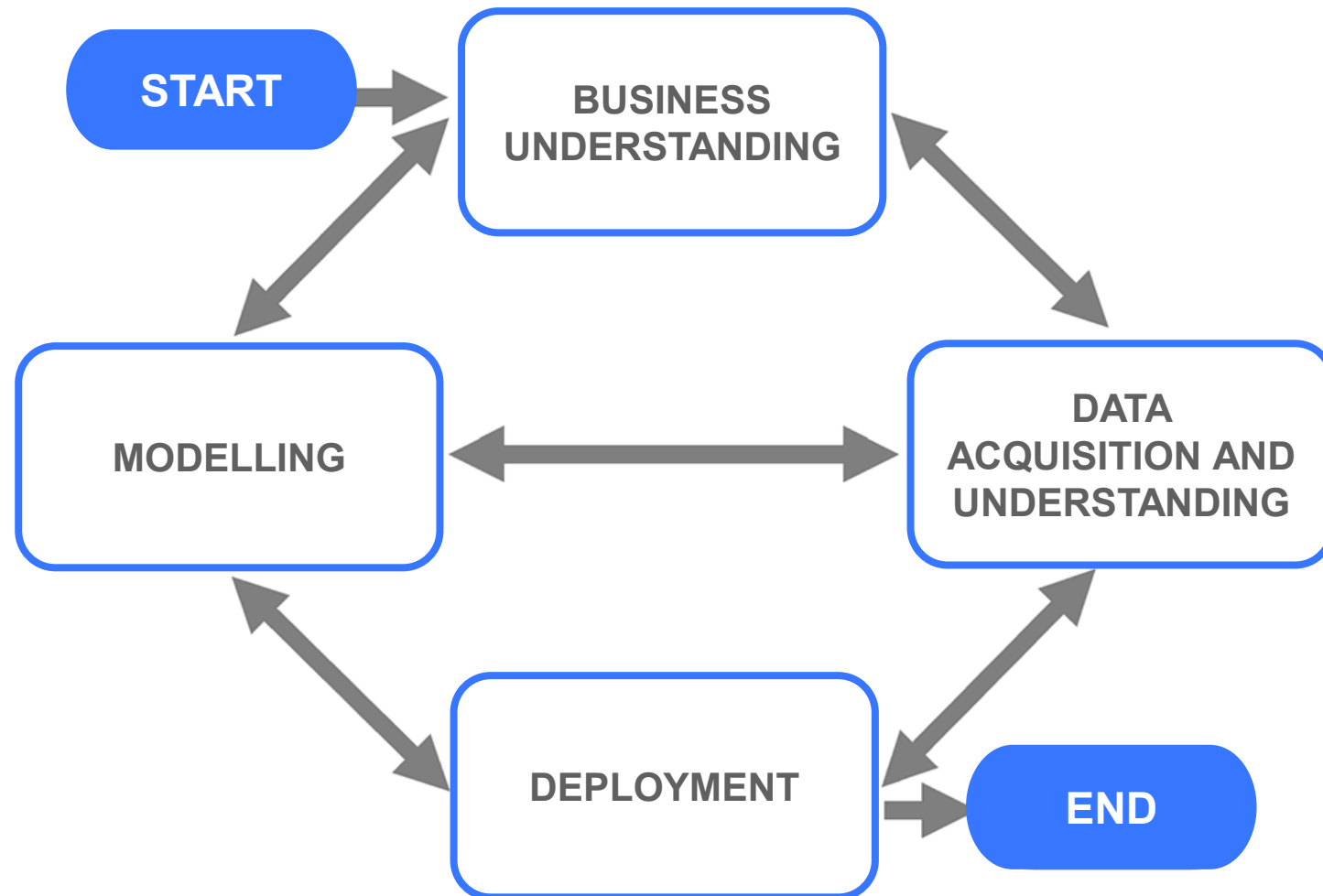
TDSP – Team Data Science Process

TDSP

- TDSP (Team Data Science Process) is a methodology developed by **Microsoft** for data science projects.
- It **includes five stages**: Business Understanding, Data Acquisition and Understanding, Modeling, Deployment, and Customer Acceptance.
- TDSP **emphasizes collaboration among team members**, scalability, and reproducibility in the data analysis process.
- Combines elements of the core data science life cycle, software engineering, and Agile processes.

3.2. Methodologies for Supervised Learning

TDSP – Team Data Science Process



3.3. Methodologies for Supervised Learning

MLOps – Machine Learning Operations

MLOps

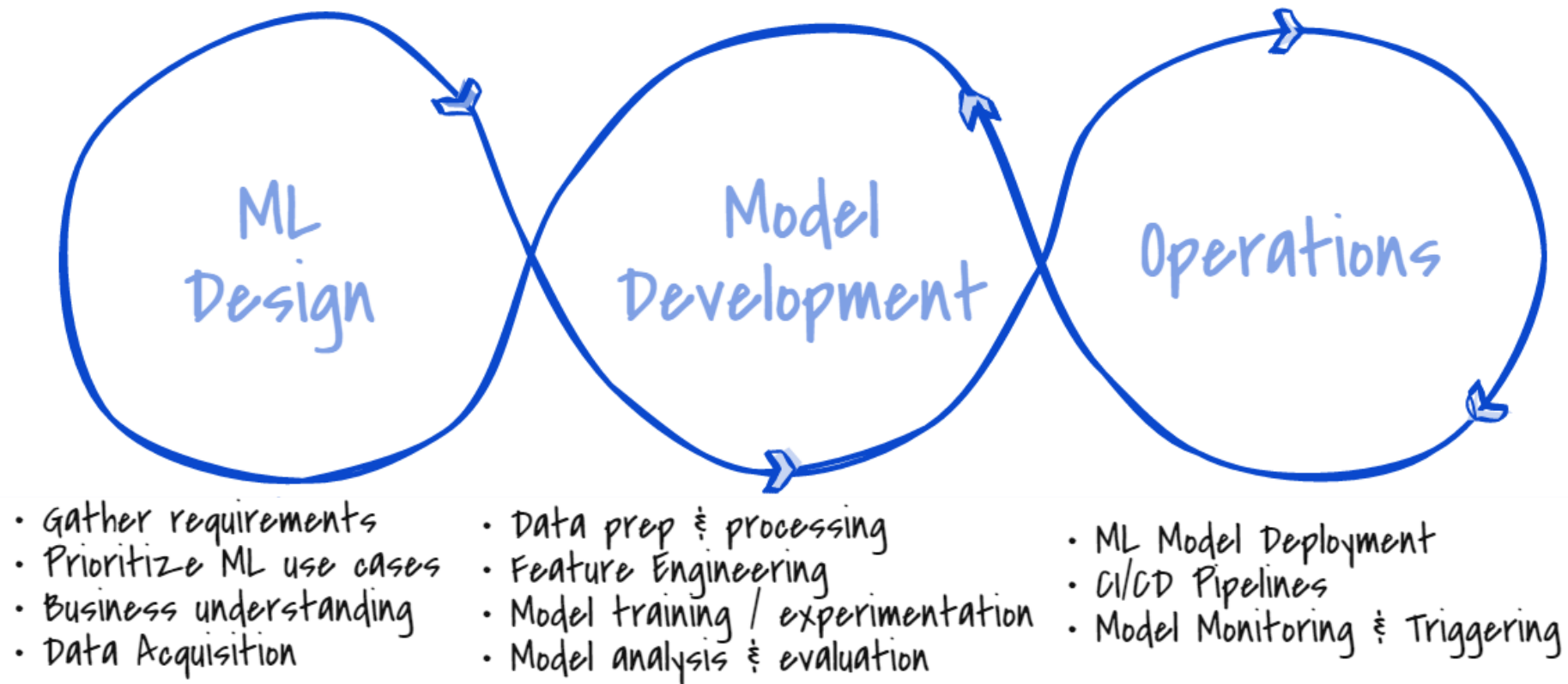
- MLOps aims to **unify** the release cycle for **machine learning and software application release**.
- MLOps **enables automated testing of machine learning artifacts** (e.g. data validation, ML model testing, and ML model integration testing)
- MLOps enables the application of **agile principles to machine learning projects**.
- ...
- MLOps must be a language-, framework-, platform-, and infrastructure-agnostic practice.

Source: [ML Ops: Machine Learning Operations](#)

3.3. Methodologies for Supervised Learning

MLOps – Machine Learning Operations

Machine Learning Operations (MLOps)

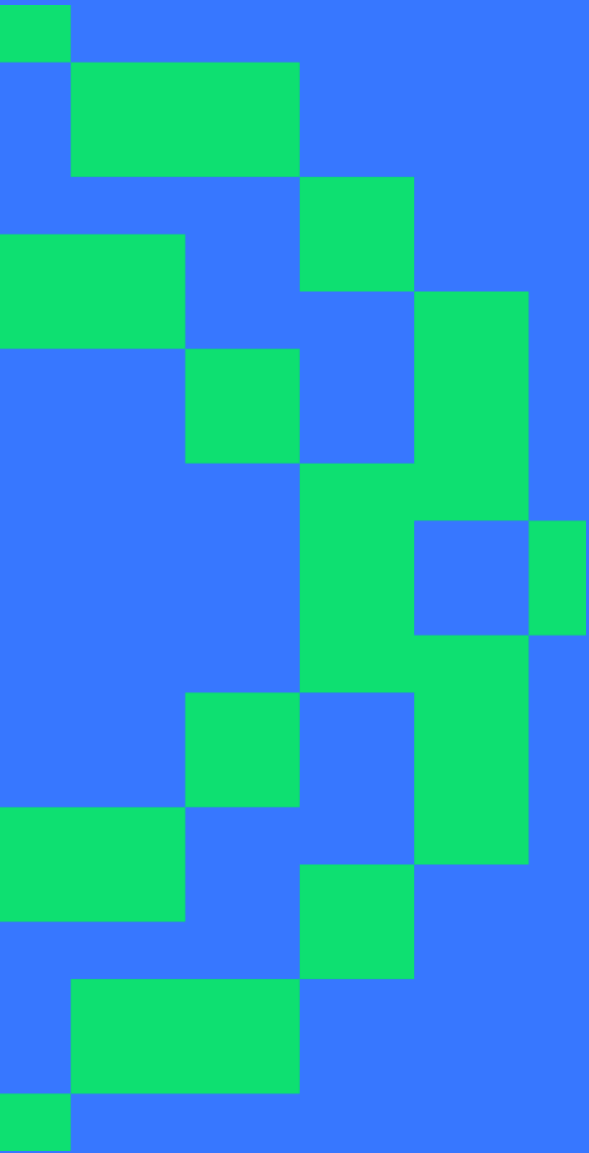


[Source: What is MLOps? An Introduction to the World of Machine Learning Operations | D-Lab](#)

3. Methodologies for Supervised Learning

RECAP

Framework	Type	Main Focus	Best For
CRISP-DM	Process model	Business understanding + structured DS workflow	Traditional analytics / DS projects
TDSP	DS lifecycle framework	Collaboration + reproducibility + standard workflow	Team-based industry DS projects
MLOps	Engineering + operations practice	Deployment + automation + monitoring + maintenance	Production ML systems at scale



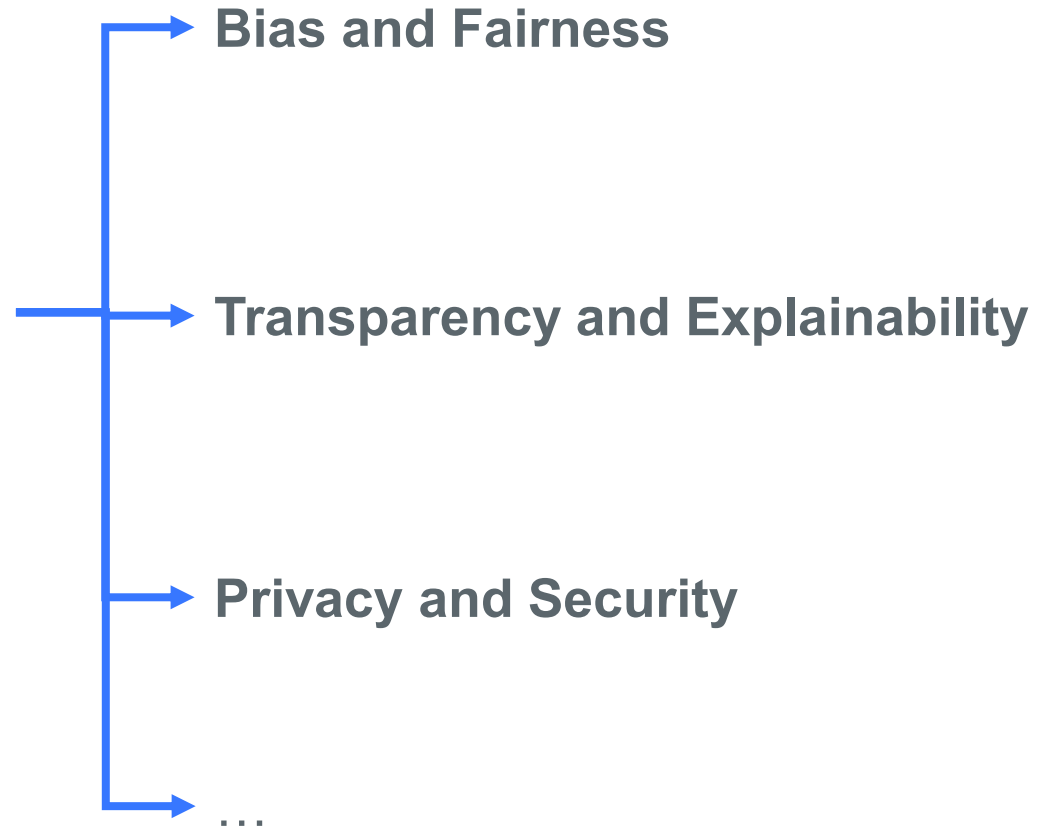
04.

Ethical Considerations

4. Ethical Considerations

Introduction

ETHICAL CONSIDERATIONS



4.1. Ethical Considerations

Bias and Fairness

Bias and Fairness – Some Examples



The first woman CEO to appear in a Google Images search is ... CEO Barbie

pcworld.com · 2015 ▾

The Ellen Pao-Kleiner Perkins trial shone a light on discrimination in the tech industry, but for a more immediate look at the challenges women face in corporate America, look no further than a Google Images search.



FaceApp apologises for 'racist' filter that lightens users' skintone

theguardian.com · 2017 ▾

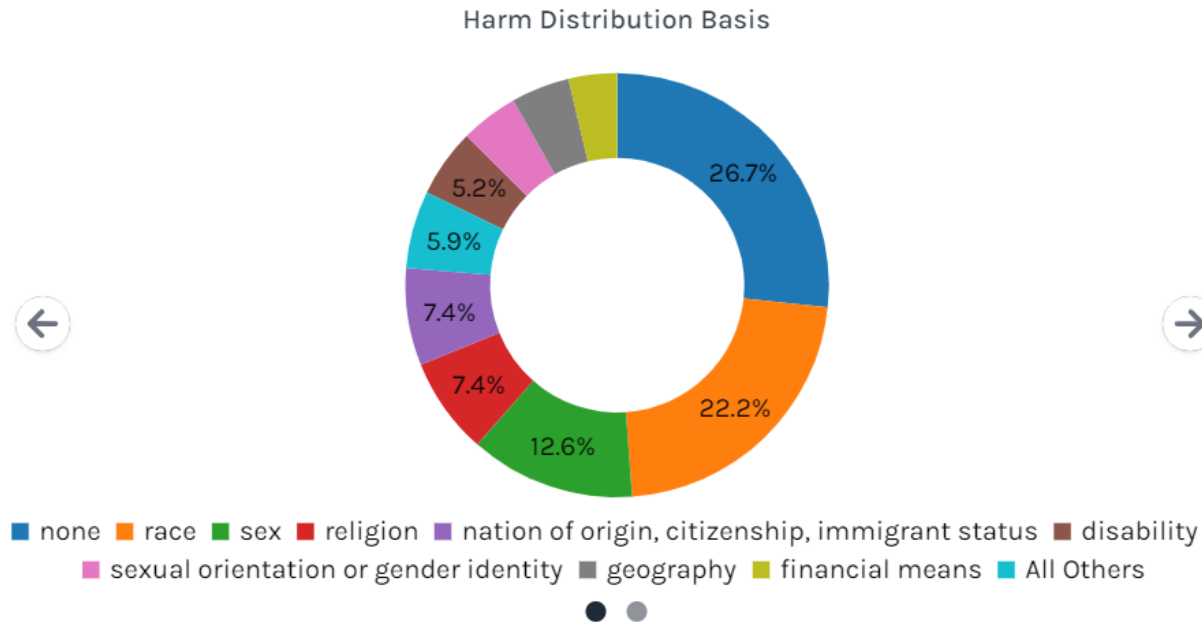
The creator of an app which changes your selfies using artificial intelligence has apologised because its “hot” filter automatically lightened people’s skin.

FaceApp is touted as an app which uses “neural networks” to change facial characte...

4.1. Ethical Considerations

Bias and Fairness

Bias and Fairness – Some Examples



How to prevent bias?

- Select training **data** that is both **comprehensive** and **sufficiently large** to mitigate prevalent biases in machine learning, such as skewed samples and prejudice bias.
- **Validate and test models rigorously** to ensure immunity against biases originating from algorithms or datasets.
- Implement **continuous monitoring** of machine learning systems during operation to avoid the gradual accumulation of biases as these systems continually learn and adapt.

4.2. Ethical Considerations

Transparency and Explainability

Transparent models

Linear Regression
Decision Trees
Logistic Regression
Naïve Bayes
...

VS

Not-so-transparent models

Neural Networks (Deep Learning)
SVM
Random Forests
Gradient Boosting
...

4.3. Ethical Considerations

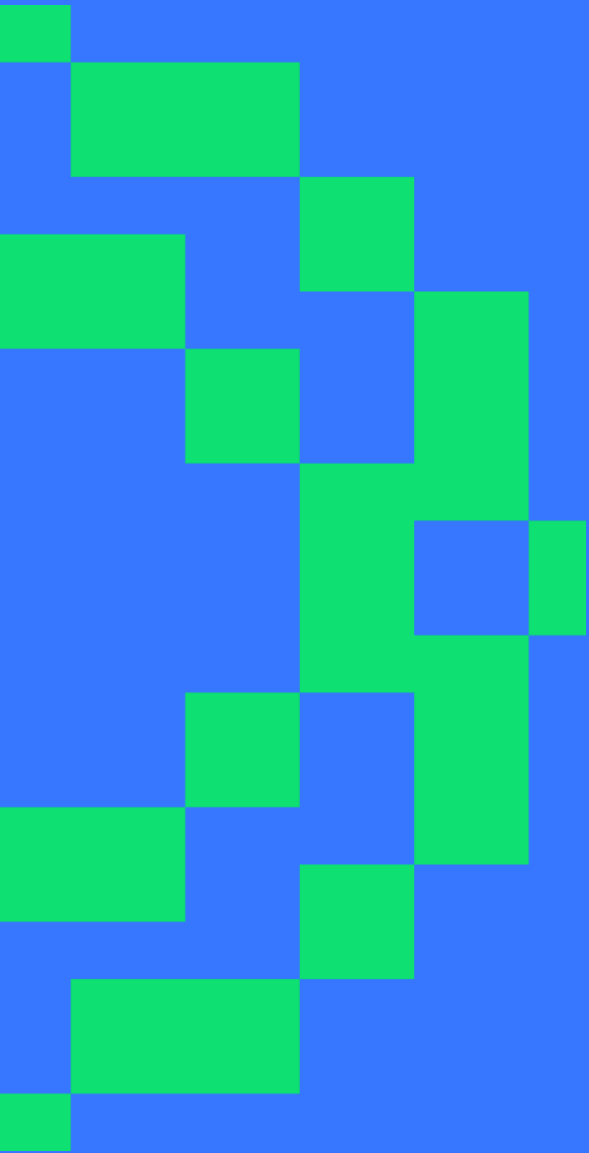
Privacy and Security

Privacy

- Data Protection Laws and Regulations (GDPR – General Data Protection Regulation)
- Anonymization
- Consent and User Control

Security

- Cybersecurity Threats
- Encryption
- Incident Response Plan
- Security Audits and Compliance
- ...



05. Challenges and Future Trends

5.1. Challenges and Future Trends

Challenges

What are the challenges?

Ethical Concerns



Managing biases, ensuring fairness, and navigating ethical dilemmas in AI applications, especially in sensitive areas like healthcare and criminal justice.

Data Quality and Accessibility



Ensuring high-quality, diverse, and accessible data for training AI models, addressing issues with biases, and dealing with data privacy regulations.

Interpretability and Explainability



Enhancing the interpretability of complex AI models to understand their decisions and ensuring transparency, especially in critical applications.

5.1. Challenges and Future Trends

Challenges

What are the challenges?

Robustness and Security



Developing AI systems that are resilient against adversarial attacks, ensuring cybersecurity, and safeguarding against data breaches and manipulation.

Talent and Expertise Gap



Addressing the shortage of skilled professionals in AI and data science, both in technical expertise and domain knowledge.

5.2. Challenges and Future Trends

Future trends

What are the future trends?

AI Ethics and Governance



Continued focus on ethical frameworks, regulations, and standards governing AI applications to ensure responsible and transparent deployment.

Explainable AI



Advancements in making AI models more interpretable, enabling users to understand and trust AI-driven decisions.

AI and automation



Further integration of AI in automation across industries and entities, driving efficiency, and streamlining processes in risk assessment, compliance monitoring, finance, and more.

5.2. Challenges and Future Trends

Future trends

What are the future trends?

Responsible AI and fairness



Continued emphasis on developing AI systems that reflect diverse perspectives, ensuring fairness, and reducing biases.

Technology adoption



The increase adoption of technology and AI, machine learning, and data analytics, to enhance oversight, risk assessment, and compliance monitoring.

Obrigada!

Thank you!

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